Zeek

- 네트워크 침입탐지시스템(NIDS)

  - Bro 또는 프로토콜 분석

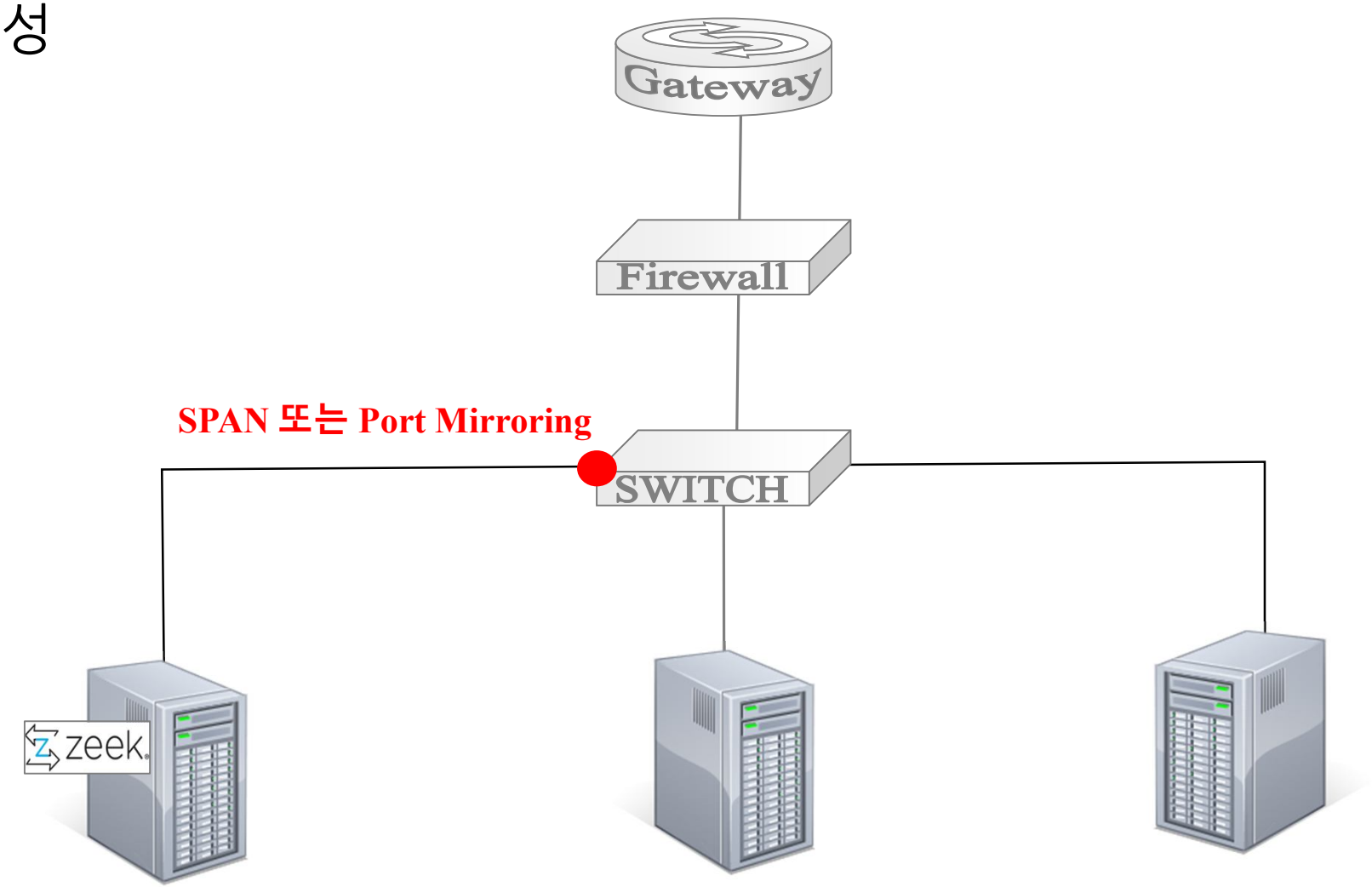
  - 네트워크를 모니터링 할 수 있는 오픈 소스 프로그램

- IP헤더와 TCP 헤더를 분석하여 로그 생성

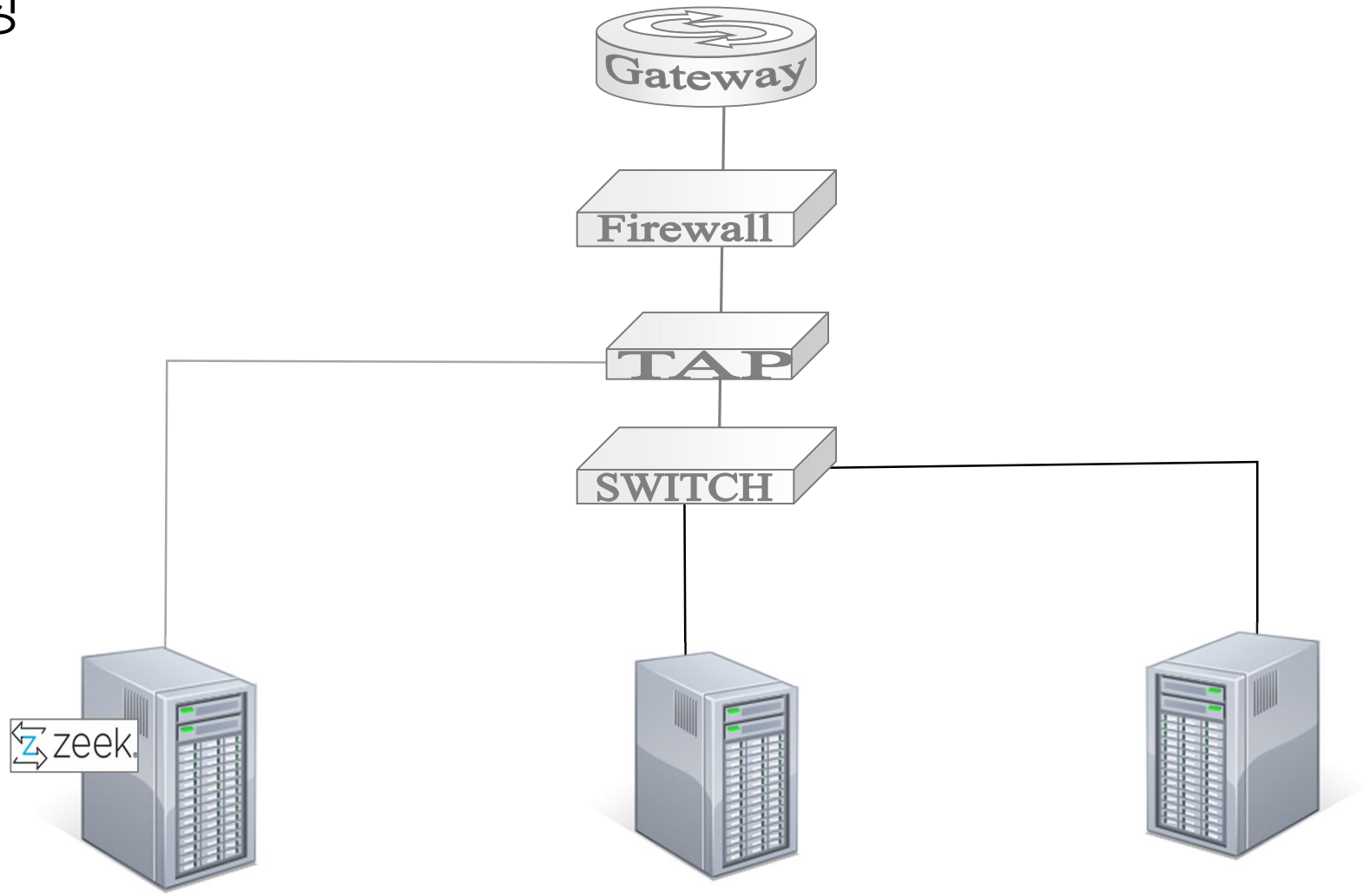
- 응용 프로토콜의 헤더를 분석하여 로그 생성

  - FTP, HTTP, SMTP, X.509 ..

Sniffing 환경구성

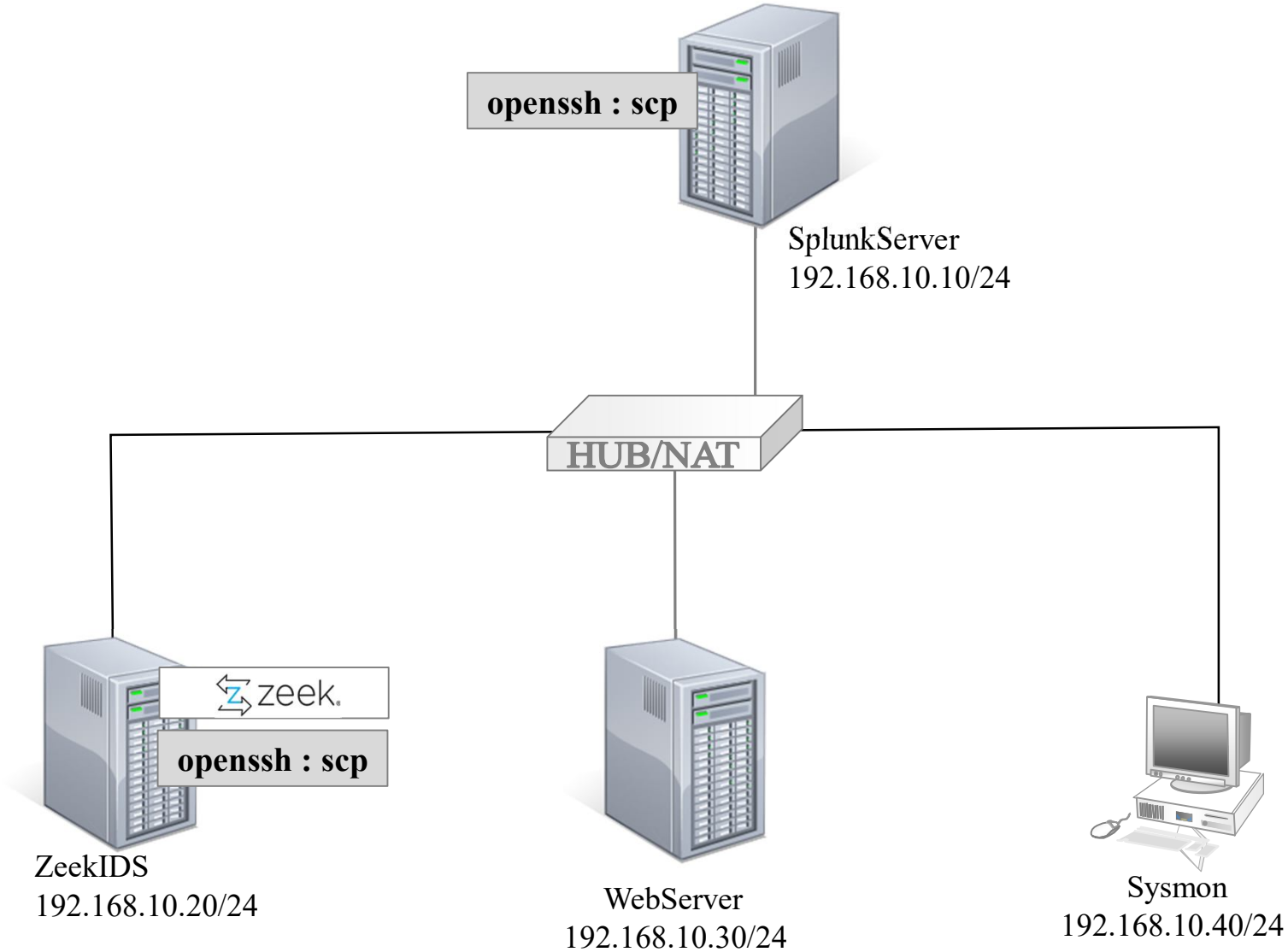


Sniffing 환경구성



Zeek Installation

1) [ZeekIDS] Zeek Installation
2) [SplunkServer] scp installation


openssh : scp
SplunkServer
192.168.10.10/24
HUB/NAT
zeek
openssh : scp
ZeekIDS
192.168.10.20/24
WebServer
192.168.10.30/24
Sysmon
192.168.10.40/24

1) [ZeekIDS] Zeek Installation

① 설치 전 환경 설정

```
#apt update
#apt-get install curl gnupg2 wget –y

#curl -fsSL https://download.opensuse.org/repositories/security:zeek/xUbuntu_24.04/Release.key | gpg --dearmor | tee /etc/apt/trusted.gpg.d/security_zeek.gpg

#echo 'deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04/ /' | tee /etc/apt/sources.list.d/security:zeek.list
```

② Zeek Installation

#apt update -y

#apt-get install zeek -y

```
Postfix Configuration
Please select the mail server configuration type that best meets your needs.

 No configuration:
  Should be chosen to leave the current configuration unchanged.
 Internet site:
  Mail is sent and received directly using SMTP.
 Internet with smarthost:
  Mail is received directly using SMTP or by running a utility such
  as fetchmail. Outgoing mail is sent using a smarthost.
 Satellite system:
  All mail is sent to another machine, called a 'smarthost', for
  delivery.
 Local only:
  The only delivered mail is the mail for local users. There is no
  network.

General mail configuration type:

                          No configuration
                          Internet Site
                          Internet with smarthost
                          Satellite system
                          Local only

                    <Ok>                        <Cancel>
```

```
Postfix Configuration
The 'mail name' is the domain name used to 'qualify' _ALL_ mail addresses without a domain name. This includes mail to and from <root>:
please do not make your machine send out mail from root@example.org unless root@example.org has told you to.

This name will also be used by other programs. It should be the single, fully qualified domain name (FQDN).

Thus, if a mail address on the local host is foo@example.org, the correct value for this option would be example.org.

System mail name:

zeekids

                                <Ok>                                <Cancel>
```

③ PATH 설정

#echo "export PATH=$PATH:/opt/zeek/bin" >> ~/.bashrc

#source ~/.bashrc

#zeek --version

```
root@zeekids:~# echo "export PATH=$PATH:/opt/zeek/bin" >> ~/.bashrc
root@zeekids:~# source ~/.bashrc
root@zeekids:~# zeek --version
zeek version 6.2.1
root@zeekids:~#
```

④ Zeek Configuration

#cd /opt/zeek/etc

#ls -l

#cat /opt/zeek/etc/networks.cfg

```
root@zeekids:/#
root@zeekids:/# cd /opt/zeek/etc
root@zeekids:/opt/zeek/etc# ls -l
total 16
-rw-rw-r-- 1 root zeek  441 Jun 30 16:50 networks.cfg
-rw-rw-r-- 1 root zeek  651 Jan 29  2015 node.cfg
-rw-rw-r-- 1 root zeek 3052 Jan 29  2015 zeekctl.cfg
drwxr-xr-x 2 root zeek 4096 Jun 30 16:44 zkg
root@zeekids:/opt/zeek/etc#
root@zeekids:/opt/zeek/etc# cat networks.cfg
# List of local networks in CIDR notation, optionally followed by a descriptive
# tag. Private address space defined by Zeek's Site::private_address_space set
# (see scripts/base/utils/site.zeek) is automatically considered local. You can
# disable this auto-inclusion by setting zeekctl's PrivateAddressSpaceIsLocal
# option to 0.
#
# Examples of valid prefixes:
#
# 1.2.3.0/24        Admin network
# 2607:f140::/32    Student network
```

#ifconfig

```
root@zeekids:/opt/zeek/etc# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.10.20  netmask 255.255.255.0  broadcast 192.168.10.255
        ether 00:0c:29:82:8d:89  txqueuelen 1000  (Ethernet)
        RX packets 73946  bytes 106906174 (106.9 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 7618  bytes 637261 (637.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

#nano /opt/zeek/etc/node.cfg

[zeek]

type=standalone

host=localhost

interface=ens33

```
root@zeekids:/opt/zeek/etc# cat node.cfg
# Example ZeekControl node configuration.
#
# This example has a standalone node ready to go except for possibly changing
# the sniffing interface.

# This is a complete standalone configuration.  Most likely you will
# only need to change the interface.
[zeek]
type=standalone
host=localhost
interface=ens33
```

⑤ Zeek 활성화

#zeekctl check

#zeekctl deploy

#zeekctl status

#ls –l /opt/zeek/spool/zeek

```
root@zeekids:/opt/zeek/etc# zeekctl check
Hint: Run the zeekctl "deploy" command to get started.
zeek scripts are ok.
root@zeekids:/opt/zeek/etc# zeekctl deploy
checking configurations ...
installing ...
creating policy directories ...
installing site policies ...
generating standalone-layout.zeek ...
generating local-networks.zeek ...
generating zeekctl-config.zeek ...
generating zeekctl-config.sh ...
stopping ...
stopping zeek ...
starting ...
starting zeek ...
root@zeekids:/opt/zeek/etc# zeekctl status
Name         Type       Host          Status    Pid    Started
zeek         standalone localhost     running   5950   30 Jun 16:55:26
root@zeekids:/opt/zeek/etc#
```

⑥ Network Log 확인

#ls –l /opt/zeek/spool/zeek

```
root@zeekids:/opt/zeek/etc# ls -l /opt/zeek/spool/zeek
total 80
-rw-r--r-- 1 root zeek   250 Jun 30 16:56 capture_loss.log
-rw-r--r-- 1 root zeek   703 Jun 30 16:57 conn.log
-rw-r--r-- 1 root zeek 33012 Jun 30 16:55 loaded_scripts.log
-rw-r--r-- 1 root zeek   753 Jun 30 16:56 notice.log
-rw-r--r-- 1 root zeek   251 Jun 30 16:55 packet_filter.log
-rw-r--r-- 1 root zeek   666 Jun 30 16:55 reporter.log
-rw-r--r-- 1 root zeek   710 Jun 30 16:55 stats.log
-rw-r--r-- 1 root zeek    20 Jun 30 16:55 stderr.log
-rw-r--r-- 1 root zeek   204 Jun 30 16:55 stdout.log
-rw-r--r-- 1 root zeek  7779 Jun 30 16:56 telemetry.log
-rw-r--r-- 1 root zeek   366 Jun 30 16:55 weird.log
root@zeekids:/opt/zeek/etc#
```

#tail /opt/zeek/spool/zeek/conn.log

```
root@zeekids:/opt/zeek/spool/zeek# tail /opt/zeek/spool/zeek/conn.log
#separator \x09
#set_separator  ,
#empty_field    (empty)
#unset_field    -
#path   conn
#open   2024-06-30-16-57-09
#fields ts      uid     id.orig_h       id.orig_p       id.resp_h       id.resp_p       proto   service duration        orig_bytes      resp_byte
s       conn_state      local_orig      local_resp      missed_bytes    history orig_pkts       orig_ip_bytes   resp_pkts       resp_ip_bytes   t
unnel_parents
#types  time    string  addr    port    addr    port    enum    string  interval        count   count   string  bool    bool    count   string  c
ount    count   count   count   set[string]
1719734166.318141       C8Mp8r1X6RuVaMosyc      192.168.10.1    49449   239.255.255.250 1900    udp     -       3.002355        700     0       S
0       T       F       0       D       4       812     0       0       -
1719734166.350914       CnBH2y1AGbXTFI2URe      192.168.10.1    49451   239.255.255.250 1900    udp     -       3.001604        700     0       S
0       T       F       0       D       4       812     0       0       -
```

2) [ZeekIDS] scp installation

- SCP를 이용하기 위해서는 SSH 서비스가 활성화 되어있어야 함

  #apt-get install –y openssh-server

  #ufw allow 22/tcp

*scp [전송파일명] [서버계정명]@[전송받을 서버]:[전송받을 서버 디렉터리]

#scp conn.log splunkserver@192.168.10.10:/tmp

```
root@zeekids:/opt/zeek/spool/zeek# scp conn.log splunk@192.168.10.10:/tmp
The authenticity of host '192.168.10.10 (192.168.10.10)' can't be established.
ED25519 key fingerprint is SHA256:ItjdbDmZN0FzdmbOTmL1sVz1bN0BBcInB3XsIGESJ7w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.10.10' (ED25519) to the list of known hosts.
splunk@192.168.10.10's password:
conn.log                                      100% 5207     3.5MB/s   00:00
root@zeekids:/opt/zeek/spool/zeek#
```

[ZeekIDS] 압축한 로그파일을 Splunk Server에 전송

#cd /opt/zeek/spool

#zip zeek.zip zeek

#scp zeek.zip splunkserver@192.168.10.10:/tmp

```
root@zeekids:/opt/zeek/spool# ls
brokerstore  installed-scripts-do-not-touch  state.db   zeek  zeek.zip  zeekctl-config.sh
root@zeekids:/opt/zeek/spool# ls -l
total 76
drwxr-xr-x 2 root root  4096 Jan 29  2015 brokerstore
drwxr-sr-x 4 root zeek  4096 Jun 30 16:55 installed-scripts-do-not-touch
-rw-r--r-- 1 root zeek 12288 Jun 30 16:55 state.db
-rw-r--r-- 1 root zeek    39 Jun 30 16:55 stats.log
drwxrws--- 2 root zeek  4096 Jun 30 17:00 tmp
drwxr-sr-x 2 root zeek  4096 Jun 30 17:11 zeek
-rw-r--r-- 1 root zeek 38817 Jun 30 17:25 zeek.zip
-rw-r--r-- 1 root zeek  2621 Jun 30 16:55 zeekctl-config.sh
root@zeekids:/opt/zeek/spool#
root@zeekids:/opt/zeek/spool# scp zeek.zip splunk@192.168.10.10:/tmp
splunk@192.168.10.10's password:
zeek.zip                                          100%   38KB  14.0MB/s   00:00
root@zeekids:/opt/zeek/spool#
```

[SplunkServer] 전송 받은 zeek.zip 압축해제 후 확인

#cd /tmp

#ls –l zeek.zip

#unzip zeek.zip

```
root@splunk:/tmp# ls zeek.zip
zeek.zip
root@splunk:/tmp#
root@splunk:/tmp# ls -l zeek.zip
-rw-r--r-- 1 splunk splunk 38817 Jun 30 17:25 zeek.zip
root@splunk:/tmp#
root@splunk:/tmp# unzip zeek.zip
Archive:  zeek.zip
   creating: zeek/
  inflating: zeek/stdout.log
  inflating: zeek/stderr.log
  inflating: zeek/.env_vars
  inflating: zeek/.cmdline
  inflating: zeek/.startup
```

[참고] HTTP 로그 필드

필드명	필드 설명
id.orig_h	송신지IP주소
id.orig_p	송신지 port번호
id.resp_h	수신지IP주소
id.resp_p	수신지 port번호
method	서비스 요청 방식
host	접근 도메인
uri	접근 상세주소(도메인제외)
referrer	Host 접속시 경유주소
version	HTTP 버전
user-agent	웹브라우저 정보

필드명	필드 설명
request_body_len	서버에게 전송하는 정보길이
response_body_len	사용자에게 전송하는 정보길이
status_code	상태코드
status_msg	상태 메시지
Proxied	프록시 접속 여부
resp_fuids	연결된 파일 ID
resp_mime_types	전송 파일 mime type

[참고] HTTP 로그 필드

#cat /tmp/http.log

```
root@splunk:/tmp/zeek# cat http.log
#separator \x09
#set_separator  ,
#empty_field    (empty)
#unset_field    -
#path   http
#open   2024-06-30-17-00-20
#fields ts      uid     id.orig_h       id.orig_p       id.resp_h       id.resp_p       trans_depth     method  host    uri     referrer        version user_agent      orig
in      request_body_len        response_body_len       status_code     status_msg      info_code       info_msg        tags    username        password        proxied orig
_fuids  orig_filenames  orig_mime_types resp_fuids      resp_filenames  resp_mime_types
#types  time    string  addr    port    addr    port    count   string  string  string  string  string  string  string  count   count   count   string  count   string  set[
enum]   string  string  set[string]     vector[string]  vector[string]  vector[string]  vector[string]  vector[string]  vector[string]
1719734420.564046       CHzhLm40k1oyqykrG9      192.168.10.20   55346   91.189.91.49    80      1       -       -       -       -       1.1     -       -       0       0  2
04      No Content      -       -       (empty) -       -       -       -       -       -       -       -       -
1719734622.787626       CAEflh3Y1r2NeHqN09      192.168.10.10   43476   91.189.91.98    80      1       GET     connectivity-check.ubuntu.com.  /       -       1.1     -  2
04      No Content      -       -       (empty) -       -       -       -       -       -       -       -       -
1719734623.330690       C1fbhP1H7kxKELJ2b8      192.168.10.10   55494   91.189.91.98    80      1       GET     connectivity-check.ubuntu.com.  /       -       1.1     -  2
04      No Content      -       -       (empty) -       -       -       -       -       -       -       -       -
1719734720.612700       ChAuaj2qtGVxsptOR5      192.168.10.20   53218   185.125.190.97  80      1       -       -       -       -       1.1     -       -       0       0  2
```